

UCS405 Discrete Mathematical Structures

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Course Objective: The course objective is to provide students with an overview of Discrete Mathematical Structures. Students will learn about topics such as logic and proofs, sets and functions, graph theory, boolean algebra, number theory and other important discrete math concepts.

Syllabus

Sets, Relations, and Functions: Sets: Operations on set, Inclusion-exclusion principle, Representation of Discrete Structures, Fuzzy set, Multi-set, bijective function, Inverse and Composition of functions, Floor and Ceiling functions, Growth of functions: Big-O notation, Big-Omega and Big-Theta Notations, Determining complexity of a program, Hash functions.

Relations: Different types of relation and their representation, Equivalence and partial-ordered relations, Partition and Covering of a set, N-ary relations and database, Closure of relations, Warshall's algorithm, Lexicographic ordering, Hasse diagram, Lattices, Boolean algebra.

Graphs Theory: Representation, Type of Graphs, Paths and Circuits: Euler Graphs, Hamiltonian Paths & Circuits; Cut-sets, Connectivity and Separability, Planar Graphs, Isomorphism, Graph Coloring, Covering and Partitioning, Application of Graph theory in real-life applications.

Basic Logic: Propositional logic, Logical connectives, Truth tables, Normal forms (conjunctive and disjunctive), Validity of well-formed formula, Propositional inference rules (concepts of modus ponens and modus tollens), Predicate logic, Universal and existential quantification, Proof Techniques.

Recurrence Relation: Solving linear recurrence relations, divide and conquer algorithms and recurrence relations.

Algebraic Structures: Group, Semi group, Monoids, Ring, Field, Homomorphism.

Number Theory: Divisibility and Modular Arithmetic, Solving Congruences, Applications of Congruences, Cryptographic applications

Laboratory Work

NA

Course Learning Objectives (CLO)

The students will be able to:

1. Perform operations on various discrete structures such as set, function, and relation.
2. Apply basic concepts of asymptotic notation in the analysis of the algorithm.
3. Illustrate the basic properties and algorithms of graphs and apply them in modelling and solving real-world problems.
4. Comprehend formal logical arguments and translate statements from a natural language into their symbolic structures in logic.

5. Identify and prove various properties of rings, fields, and groups.
6. Illustrate and apply the division algorithm, mod function, and Congruence.

Text Books

1. Discrete Mathematics and its Applications, Rosen H. K., McGraw Hill, 7th ed., 2011
2. Discrete Mathematical Structures with Applications to Computer Science, Tremblay P. J. and Manohar, R., Tata McGraw Hill, 2008.

Reference Books

1. Contemporary Abstract Algebra, Gallian A. J., Cengage Learning, 9th ed., 2017
2. Discrete Mathematics, Lipschutz S., Lipson M., McGraw-Hill, 3rd ed., 2007